

Programming Contest Report

1. Introduction

This report presents an analysis of student performance in a Python programming contest consisting of four problems:

- Looping with Number
- Fibonacci Numbers
- Array Sum
- Reverse a List

Each problem carried partial scores, and students attempted all problems during the contest. The aim of this report is to evaluate overall performance, identify trends, highlight top performers, and extract insights for future training.

2. Course Assessment Analysis

- Overall performance shows **moderate to low success rates**.
- Most students scored **0 in multiple problems**, indicating gaps in fundamental understanding.
- Only a **small group of students consistently scored across multiple problems**.
- Highest observed scores:
 - Fibonacci: up to **15**
 - Array Sum: up to **10**
 - Looping: up to **10**
- The **Reverse a List** problem had **almost universally poor performance**, suggesting it was the most difficult or least understood.

3. Students Who Performed Well

The following students demonstrated strong performance across multiple problems:

Top Performers

- **Raghavendra** – High scores in Looping, Fibonacci, Array Sum
- **Rohith** – Strong consistent performance (10s in 3 problems)
- **Kiran** – Performed well in 3 problems
- **Manjunath** – Good performance across 3 problems
- **Bhavana** – Strong in Fibonacci and Array Sum

- **Ruthwik** – Good consistency
- **Yogesh** – Balanced performance
- **Deepak** – Highest Fibonacci score (15)

Moderate Performers

- Shalini
- Prateek
- Ganavi
- Lishmitha

These students showed partial understanding but lacked consistency.

4. Problem-wise Performance Analysis

1. Looping with Number

- Moderate performance overall
- Several students scored **5 or 10**
- Indicates **basic loop understanding exists**, but not universal

2. Fibonacci Numbers

- Mixed performance
- Some high scorers (10–15), but many zeros
- Suggests difficulty in **logic building and sequence generation**

3. Array Sum

- Better performance compared to others
- Many students scored partial/full marks
- Indicates **comfort with arrays and basic operations**

4. Reverse a List

- **Worst-performing problem**
- Almost all students scored **0**
- Indicates major gaps in:
 - List manipulation
 - Indexing
 - Python-specific methods

5. Problem-Specific Analysis

Problem	Difficulty Level	Observations
Looping	Easy–Medium	Basic concept understood by some
Fibonacci	Medium	Logic-heavy; many struggled
Array Sum	Easy	Most approachable problem
Reverse List	Medium	Poor performance; conceptual gap

6. Common Error Types

From the pattern of scores, likely errors include:

- **Syntax errors**
- Incorrect loop usage
- Poor understanding of **iteration boundaries**
- Mistakes in **index handling**
- Incomplete logic implementation
- Lack of familiarity with Python list operations
- Time management issues (not reaching later problems)

7. Insights from Training Perspective

Key Learning Gaps

- Weak fundamentals in **loops and sequences**
- Poor understanding of **list manipulation**
- Difficulty in **problem-solving logic**

Recommendations

- Conduct **hands-on coding sessions**
- Focus on:
 - Looping patterns
 - Fibonacci logic
 - List operations (reverse, slicing)
- Introduce **step-by-step problem solving**
- Encourage **practice on basic problems**
- Use **debugging exercises**

8. Contest Insights

- Many students attempted but could not score → **engagement is high, but skill level needs improvement**
- Performance drops significantly after basic problems
- Reverse List problem acted as a **performance separator**
- A small group of students shows **strong potential for advanced training**